

Final Exam Statistical Mechanics

10 January 2025

Welcome to the final exam of Statistical Mechanics of the GEMIF. The exam covers all the topics we have seen during the course.

The final mark for the course will be $= \max\{0.40 \text{ midterm} + 0.60 \text{ final}, 1.00 \text{ final}\}$

Some useful relations

$$\int_0^\infty z^{-3/4} e^{-z} dz = \Gamma\left(\frac{1}{4}\right).$$

Question 1 [1.5 points]

Let X and Y be independent continuous random variables uniformly distributed in the interval $[0, 1]$. Find the probability density function (PDF) of $W = X + Y$ and plot it.

Solution Q1

$$\begin{aligned} f_W(w) &= \lim_{\delta \rightarrow 0} P(w \leq X + Y \leq w + \delta) = \int_{-\infty}^{\infty} \int_{w-x}^{w-x+\delta} f_X(x) f_Y(y) dy dx \\ &\approx \int_{-\infty}^{\infty} f_X(x) f_Y(w-x) \delta dx. \end{aligned}$$

The integrand $f_X(x) f_Y(w-x)$ is nonzero (and equal to 1) for $0 \leq x \leq 1$ and $0 \leq w-x \leq 1$. Combining these two inequalities, the integrand is nonzero for:

$$\max\{0, w-1\} \leq x \leq \min\{1, w\}.$$

Thus,

$$f_W(w) = \begin{cases} \min\{1, w\} - \max\{0, w-1\}, & 0 \leq w \leq 2, \\ 0, & \text{otherwise.} \end{cases}$$

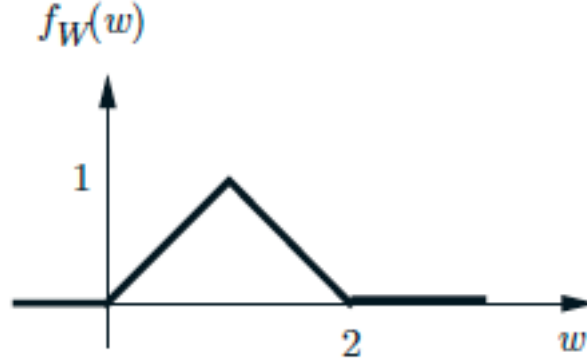


Figure 1: Plot of the PDF of $W = X + Y$.

Question 2 [1 point]

Consider a lattice with M sites and n particles that can be either blue (B) or red (R) occupying sites in the lattice (M and n are fixed). Sites occupied with red particles have energy ϵ , and sites occupied with blue particles have energy 2ϵ . Empty sites have energy 0. Find the entropy of the system in the microcanonical ensemble in the thermodynamic limit as a function of n , n_R and M assuming that $M \gg n_R + n_B$ (n_R , n_B are the number of red and blue particles, respectively).

Solution Question 2

In the microcanonical ensemble, we have to count the number of configurations with fixed energy. The total energy is $E = n_R \epsilon + n_B 2\epsilon$. Since $n = n_R + n_B$, we can express the energy as $E = n \epsilon + (n - n_R) 2\epsilon = (2n - n_R)\epsilon$. Therefore, configurations with the same energy have a specific number of red balls. Therefore, for each combination of n occupied sites, the number of combinations with n_R balls is $\binom{n}{n_R}$. Because the number of combination of n occupied sites in a lattice of M sites is $\binom{M}{n}$ the total number of microstates with the same energy is

$$\Omega(n_R, M, n) = \binom{M}{n} \binom{n}{n_R} = \frac{M!}{(M-n)!(n-n_R)!n_R!} \quad (1)$$

The entropy is $S = k_B \log \Omega$ which using Stirling's approximation yields in the thermodynamic limit:

$$S = -k_B M \left(\left(1 - \frac{M}{n}\right) \log\left(1 - \frac{M}{n}\right) + \frac{n_R}{M} \log \frac{n_R}{M} + \frac{n - n_R}{M} \log \frac{n - n_R}{M} \right) \quad (2)$$

Question 3 [1.5 points]

Consider an ideal gas of N fermions of mass m and Fermi energy ϵ_F at zero temperature with density of states:

$$g(\epsilon) d\epsilon = C \sqrt{\epsilon} d\epsilon. \quad (3)$$

- (a) Compute the value of the constant C .
- (b) Compute $\langle v_x^2 \rangle$.
- (c) Discuss whether the equipartition theorem holds.

Solution Q3

- (a) To compute C , we note that at zero temperature only energy levels between 0 (the minimum energy for a free gas) and the Fermi energy are occupied. Therefore, at zero temperature the number of fermions N has to be equal to the number of states below ϵ_F so that:

$$N = \int_0^{\epsilon_F} d\epsilon g(\epsilon) = \frac{C}{3/2} \epsilon_F^{3/2} \quad \rightarrow \quad C = \frac{3N}{2\epsilon_F^{3/2}} \quad (4)$$

- (b) To compute $\langle v_x^2 \rangle$ we realize that the energy of each fermion in the gas is equal to its kinetic energy so that $\langle E \rangle = N \frac{1}{2} m \langle v^2 \rangle$ with $\langle v^2 \rangle = \langle v_x^2 + v_y^2 + v_z^2 \rangle = 3\langle v_x^2 \rangle$ so that

$$\langle E \rangle = \int_0^{\epsilon_F} d\epsilon g(\epsilon) \epsilon = \frac{2N}{3\epsilon_F^{3/2}} \frac{\epsilon_F^{5/2}}{5/2} \quad \rightarrow \quad \langle v_x^2 \rangle = \frac{2\langle E \rangle}{3mN} = \frac{2\epsilon_F}{5m} \quad (5)$$

- (c) The equipartition theorem does not hold because at zero temperature the Energy would be zero and it is not.

Question 4 [1 point]

Justify that a system of N one-dimensional, classical, and independent oscillators, subjected to an anharmonic potential of the type

$$V(x) = \frac{1}{4} K x^4,$$

has a heat capacity at constant volume given by

$$C_V = \frac{3}{4} N k_B.$$

Solution Q4

The classical Hamiltonian for N particles is:

$$H = \sum_{i=1}^N \left(\frac{p_i^2}{2m} + \frac{1}{4} K x_i^4 \right).$$

We are interested in finding the expected value of H :

$$U = \langle H \rangle = \sum_{i=1}^N \left(\frac{\langle p_i^2 \rangle}{2m} + \frac{1}{4} K \langle x_i^4 \rangle \right). \quad (1)$$

The simplest way to solve this problem is to apply the generalized equipartition theorem to each variable x_i . We obtain:

$$\left\langle x_i \frac{\partial H}{\partial x_i} \right\rangle = k_B T \quad \Rightarrow \quad \langle x_i K x_i^3 \rangle = k_B T \quad \Rightarrow \quad \langle x_i^4 \rangle = \frac{k_B T}{K}. \quad (2)$$

Similarly, applying the theorem to each p_i , we have:

$$\left\langle p_i \frac{\partial H}{\partial p_i} \right\rangle = k_B T \quad \Rightarrow \quad \left\langle p_i \frac{p_i}{m} \right\rangle = k_B T \quad \Rightarrow \quad \langle p_i^2 \rangle = m k_B T. \quad (3)$$

Substituting the results from (2) and (3) into equation (1), we get:

$$U = \sum_{i=1}^N \left(\frac{m k_B T}{2m} + \frac{1}{4} K \frac{k_B T}{K} \right) = N k_B T \left(\frac{1}{2} + \frac{1}{4} \right) = \frac{3}{4} N k_B T.$$

Therefore, the heat capacity is given by:

$$C_v = \left(\frac{\partial U}{\partial T} \right)_{N,V} = \frac{3}{4} N k_B.$$

An **alternative**, but more computationally expensive, approach involves using the canonical formalism for a system of N independent particles. Here it is: the single-particle partition function is:

$$Q_1 = \int_{-\infty}^{\infty} dp e^{-\frac{\beta}{2m} p^2} \int_{-\infty}^{\infty} dx e^{-\frac{\beta k}{4} x^4}.$$

Using the symmetry of the second integral, this becomes:

$$Q_1 = \frac{1}{2} \sqrt{\frac{2m\pi}{\beta}} 2 \int_0^{\infty} dx e^{-\frac{\beta k}{4} x^4}.$$

To evaluate the integral, we perform a change of variables:

$$z = \frac{\beta k}{4} x^4 \quad \Rightarrow \quad x = \left(\frac{4z}{\beta k} \right)^{1/4}, \quad dx = \frac{dz}{(4z)^{3/4} (\beta k)^{1/4}}.$$

Substituting into the integral, we get:

$$Q_1 = \sqrt{\frac{2m\pi}{\beta}} \frac{1}{4^{3/4}(\beta k)^{1/4}} \int_0^\infty z^{-3/4} e^{-z} dz.$$

From integral tables, we find:

$$\int_0^\infty z^{-3/4} e^{-z} dz = \Gamma\left(\frac{1}{4}\right).$$

Thus, the single-particle partition function becomes:

$$Q_1 = C(k_B T)^{3/4},$$

where C is a constant that incorporates all factors outside $T^{3/4}$.

Now, following the canonical formalism, we have:

$$F = -k_B T N \ln Q_1 = -k_B T N \ln \left(C(k_B T)^{3/4} \right),$$

$$S = - \left(\frac{\partial F}{\partial T} \right)_{N,V} = k_B N \ln Q_1 + k_B T N \frac{C k_B^{\frac{3}{4}} (k_B T)^{-1/4}}{C (k_B T)^{3/4}}$$

$$U = F + TS = k_B T N \frac{C k_B^{\frac{3}{4}} (k_B T)^{-1/4}}{C (k_B T)^{3/4}} = \frac{3}{4} N k_B T.$$

Finally, the heat capacity is:

$$C_V = \frac{3}{4} N k_B.$$

Problem 1 [2.5 points]

At very high temperature a gas contained in a volume V comprises exclusively non-interacting atoms of mass m and chemical potential μ_a . At lower temperatures some of these atoms combine into diatomic molecules of mass $2m$, chemical potential μ_m and bond energy γ . Assume that the energy of each one of these molecules is their kinetic energy plus the bond energy γ .

- Compute the grand partition function of the system at high temperatures.
- Compute the grand partition function of the mixture of atoms and molecules.
- Find the expected number of atoms $\langle N_a \rangle$ and molecules $\langle N_m \rangle$. At what temperature is the expected number of atoms that are free and in diatomic molecules the same?

Solution Problem 1

- (a) At high temperature we have a standard ideal gas as we have seen in class. The grand partition function is

$$\mathcal{Z} = \sum_{n=0}^{\infty} e^{\beta\mu_a n} \frac{Z_1^n}{n!} = \exp z_a Z_1 \quad z_a = e^{\beta\mu_a} \quad Z_1 = \frac{V}{h^3} (2\pi m k_B T)^{3/2} \quad (6)$$

- (b) When we have free atoms and diatomic molecules then we have that

$$\mathcal{Z} = \sum_{n_a, n_m} e^{\beta\mu_a n_a} \frac{Z_1^{n_a}}{n_a!} e^{\beta(\mu_m - \gamma)n_m} \frac{(2^{3/2} Z_1)^{n_m}}{n_m!} = \exp(z_a Z_1) \exp(z_m Z_1 2^{3/2}) \quad (7)$$

with $z_m = \exp(\beta(\mu - \gamma))$.

- (c) We can obtain the expected number of atoms and molecules as:

$$\langle N_a \rangle = \frac{1}{\beta} \frac{\partial \log \mathcal{Z}}{\partial \mu_a} = z_a Z_1 \quad (8)$$

$$\langle N_m \rangle = \frac{1}{\beta} \frac{\partial \log \mathcal{Z}}{\partial \mu_m} = z_m Z_1 2^{3/2} \quad (9)$$

$$(10)$$

If the number of free-atoms and atoms in the molecules is the same, since there are two atoms in each molecule, the condition that needs to be fulfilled is

$$\langle N_a \rangle = 2 \langle N_m \rangle . \quad (11)$$

So that

$$z_a Z_1 = 2 z_m Z_1 2^{3/2} \rightarrow \mu_a - \mu_m + \gamma = k_B T^* \log 2^{5/2} \rightarrow T^* = \frac{\mu_a - \mu_m + \gamma}{\frac{5}{2} k_B \log 2} . \quad (12)$$

Problem 2 [2.5 point]

We consider a periodic lattice gas with N sites, where each site i can host at most one atom. The occupation number $n_i \in \{0, 1\}$ represents whether the site i is empty ($n_i = 0$) or occupied ($n_i = 1$). The Hamiltonian of the system is:

$$H_{LG} = -U \sum_{\langle i, j \rangle} n_i n_j - \mu \sum_i n_i,$$

where: $\langle i, j \rangle$ indicates summation over all nearest-neighbor pairs, $U > 0$ is the energy parameter representing attraction between atoms at neighboring sites, and the second term is the chemical potential term $-\mu \sum_i n_i = -\mu N_{\text{part}}$ of the grand canonical ensemble, with N_{part} the number of particles in the system.

- (a) Map the Hamiltonian of the lattice gas H_{LG} into the Hamiltonian of an Ising model.
- (b) Knowing the phase transition of the Ising model for the external field $h = 0$, do you expect any phase transition for the lattice gas? Discuss your reasoning.
- (c) Assuming a Mean-Field Theory (MFT) approach and $h = 0$, what is the value of the critical temperature T_c for the lattice gas?
- (c) Plot and discuss the phase diagram in the (μ, T) -plane for the lattice gas, considering that crossing the coexistence line by increasing h in the Ising model results in a jump in the magnetization from $-m_0$ to $+m_0$.

Hint: It can be useful to convert the sum over the bonds into a sum over the sites:

$$\sum_{\langle i,j \rangle} = \frac{1}{2} \sum_i \sum_{j \in v(i)},$$

where $v(i)$ is the set of q neighbors of the site i . On a regular lattice, the number of bonds is therefore linked to the number of sites:

$$N_{\text{bonds}} = \frac{Nq}{2}.$$

Solution P2

- (a) The Hamiltonian for the lattice gas is:

$$H_{\text{LG}} = -U \sum_{\langle ij \rangle} n_i n_j - \mu \sum_i n_i, \quad (4.6.2)$$

We can now map the lattice gas model onto the Ising model. Let:

$$s_i = 2n_i - 1 = \pm 1, \quad n_i = \frac{s_i + 1}{2}. \quad (4.6.3, 4.6.4)$$

Substituting these into the Hamiltonian gives:

$$H_{\text{LG}} = -U \sum_{\langle ij \rangle} \left(\frac{s_i + 1}{2} \right) \left(\frac{s_j + 1}{2} \right) - \mu \sum_i \left(\frac{s_i + 1}{2} \right). \quad (4.6.5)$$

Expanding this expression:

$$H_{\text{LG}} = -\frac{U}{4} \sum_{\langle ij \rangle} (s_i s_j + s_i + s_j + 1) - \frac{\mu}{2} \sum_i (s_i + 1). \quad (4.6.6)$$

Rearranging terms:

$$H_{\text{LG}} = -\frac{U}{4} \sum_{\langle ij \rangle} s_i s_j - \left(\frac{Uq}{2} + \frac{\mu}{2} \right) \sum_i s_i - \left(\frac{qU}{8} + \frac{\mu}{2} \right) N,$$

where q is the number of neighbors (coordination number of the lattice). Note, N is the number of lattice sites, not the number of particles.

Recalling the Hamiltonian of the Ising model with external field:

$$H_{\text{Ising}} = -J \sum_{\langle ij \rangle} s_i s_j - h \sum_i s_i,$$

we observe that the

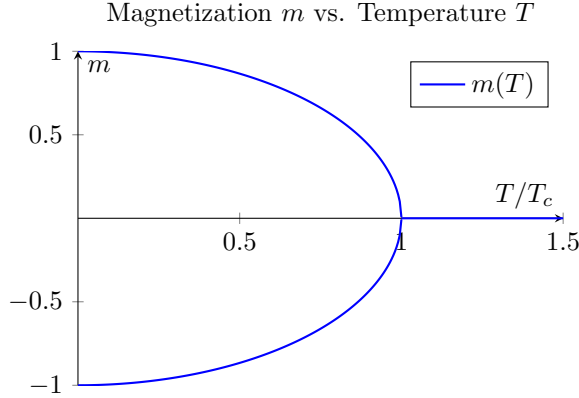
$$H_{\text{LG}} = H_{\text{Ising}} - \left(\frac{qU}{8} + \frac{\mu}{2} \right) N$$

with

$$J = \frac{U}{4}, \quad h = \frac{Uq + \mu}{2}.$$

Note that the last term is a constant (a shift of the global energy) independent of the variables of the Hamiltonian, and that can be ignored.

- (b) In the Ising model, there is a first-order coexistence line at $0 \leq T \leq T_c$ for $h = 0$, which ends at the critical endpoint $T = T_c$, $h = 0$, (remember, see figure).



Given the former mapping, the lattice gas model has a first-order coexistence line at $0 \leq T \leq T_c$ for $\mu = -qU$ (such that $h = 0$), ending at a critical endpoint at $T = T_c$, $\mu = -qU$.

- (c) In the mean-field solution of the Ising model, $k_B T_c = qJ/2$, so in the lattice gas model, $k_B T_c = qU/8$. Consequently, the critical temperature $T_c = qU/8k_B$

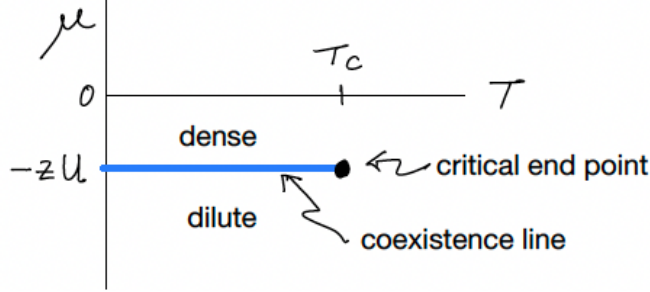


Figure 2: Plot of the phase diagram $\mu - T$.

- (d) When crossing the coexistence line by increasing h in the Ising model, there is a jump in magnetization from $-m_0$ to $+m_0$. Correspondingly, in the lattice gas model, as μ increases across the coexistence line, there is a jump in particle density:

$$n = \frac{1}{N} \sum_i \langle n_i \rangle = \langle n_i \rangle = \frac{\langle s_i \rangle + 1}{2},$$

from $n_- = \frac{-m_0+1}{2}$ to $n_+ = \frac{+m_0+1}{2}$, with a discontinuous jump $\Delta n = n_+ - n_- = m_0$. So, the plot has to be as in Figure 2.

As $T \rightarrow T_c$ along the coexistence line, the jump vanishes as $\Delta n \sim (T_c - T)^\beta$. Along the coexistence line, the average particle density is $\bar{n} = \frac{n_+ + n_-}{2} = \frac{1}{2}$. In Landau theory, $n - \bar{n}$ is the order parameter.

Thus, the lattice gas model exhibits a liquid-to-gas phase transition, where the particle density jumps discontinuously across the coexistence line. The coexistence line ends at a critical point. For $T > T_c$, there is no distinction between liquid and gas phases, and particle density varies continuously with μ . The critical behavior of the liquid-gas endpoint is identical to that of the Ising model, including critical exponents like β .